Regression Algorithms for Learning & Its Application on Crypto Currency Markets

Report 3

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1. Abstract

This report contains the computational and mathematical investigation of Ridge Regression following on from the theoretical foundations established in Reports 1 and 2. The first part of the report has a fully handworked example using a small dataset to show how the Ridge estimator is calculated step by step. The second part evaluates the behaviour of Ridge Regression within the project's proof-of-concept Python program, which applies both a closed-form self implementation and scikit-learn's Ridge model to a synthetic dataset. By comparing the results across these two contexts, the report demonstrates that both implementations give us the expected shrinkage behaviour of Ridge Regression and produce predictions consistent with the underlying model structure. The findings confirm that the theoretical formulation and practical implementation of Ridge Regression within this project are aligned thus forming a solid foundation for further model development and experimentation in later stages of the project.

2. Introduction

This report focuses on connecting the theoretical development of Ridge Regression from the previous reports with practical applications through an explicit handworked step by step numerical example and computational verification. Report 1 introduced essential regression concepts and the motivation for regularisation, while Report 2 showed previous implementations of Ridge regression concepts on financial markets. Building on this foundation, this report provides both a manually worked example of Ridge Regression and a detailed evaluation of its implementation in the proof-of-concept program developed for this project. The manual example illustrates how the penalty term modifies the solution in a simple setting, whereas the program demonstrates how Ridge Regression behaves when applied to a larger dataset with multiple predictors and noise. The comparison between these two settings serves to validate the correctness of the implementation and deepen understanding of the algorithm's behaviour.

3. Manual Handworked Example of Ridge Regression

To demonstrate Ridge Regression in its simplest form, a small dataset with one predictor variable and three observations was picked:

$$x = [1, 2, 3] \quad , \quad y = [2, 3, 6]$$

The regression model takes the usual linear form:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Ridge Regression requires forming the design matrix X and the y:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \qquad y = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$$

The Ridge estimator is defined by the expression:

$$\beta_{Ridge} = (X^T X + \lambda I)^{-1} X^T y$$

A regularisation strength of λ=1 is used to demonstrate shrinkage while also keeping the calculations simple.

The matrix is $X^T X$ calculated:

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1+1+1 & 1+2+3 \\ 1+2+3 & 1+4+9 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

The penalty term λI is calculated:

$$\lambda I = 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\lambda I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then you add the two together:

$$X^T X + \lambda I = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 4 & 6 \\ 6 & 15 \end{bmatrix}$$

Then you find the vector $X^T y$:

$$X^T y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 2+3+6 \\ 2+6+18 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 11 \\ 26 \end{bmatrix}$$

Then you inverse the matrix $(X^T X + \lambda I)$:

$$(X^T X + \lambda I)^{-1} = \frac{1}{4(15) - 6(6)} \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix} = \frac{1}{60-36} \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix}$$

$$(X^T X + \lambda I)^{-1} = \frac{1}{24} \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix}$$

Then you multiply inverse the matrix $(X^T X + \lambda I)$ by $X^T y$:

$$(X^T X + \lambda I)^{-1} (X^T y) = \frac{1}{24} \begin{bmatrix} 15 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ 26 \end{bmatrix}$$

$$(X^T X + \lambda I)^{-1} (X^T y) = \frac{1}{24} \begin{bmatrix} 165-156 \\ -66+104 \end{bmatrix} = \frac{1}{24} \begin{bmatrix} 9 \\ 38 \end{bmatrix}$$

Then we get the Ridge Coefficients:

$$\beta_0 = \frac{9}{24} = \boxed{0.375}$$

$$\beta_1 = \frac{38}{24} = \boxed{1.583}$$

We can plug them back into the model and get:

$$y = 0.375 + 1.583x + \varepsilon$$

These values confirm the following expected behaviors:

- The slope remains positive thus reflecting the relationship in the data while the penalty reduces its magnitude compared to the unregularised least squares estimate.
- Similarly the intercept is also adjusted thus demonstrating how the penalty term influences the overall model structure.

4. Application Using the Proof-of-Concept Program

While the manual example provides transparency into the mathematical procedure, the next step is to verify the behaviour of Ridge Regression through the project's proof-of-concept Python program. The program generates a synthetic dataset of 240 samples. Each sample contains three predictor variables drawn from a normal distribution. The target variable is produced using the true underlying model:

$$y = 4.6x_1 - 5.1x_2 + 2.7x_3 + 0.75 + \varepsilon$$

Noise is added to simulate real-world conditions. Before model fitting, the program standardises all predictors using StandardScaler in order to ensure equal influence of the penalty across variables.
To validate the Ridge Regression implementation, the program applies the algorithm in two independent ways:

- A closed-form ridge regression self implementation
- Scikit-learn's Ridge model

Both models are trained using the same data and the same regularisation parameter, λ=1.0. After fitting, the program generates predictions on the test set and visualises the results through 4 plots

- Alpha vs Coefficient Value for Closed Ridge
- True vs Predicted Graph for OLS
- True vs Predicted Graph for Closed Ridge
- True vs Predicted Graph for Sklearn Ridge

Both implementations produce predictions that lie close to the diagonal reference line. This demonstrates that the Ridge models successfully capture the underlying structure of the data and generalise well. The coefficient comparison plot shows that the Ridge coefficients are consistently smaller in magnitude than the unregularised coefficients. This visual confirmation strongly supports the theoretical expectation that the regularisation term shrinks coefficients towards zero. Furthermore, the closed-form and scikit-learn implementations show similar shrinkage patterns thus validating that the mathematical formula and its program implementation are functioning correctly.

5. Comparison

The handworked example and the computational implementation reinforce each other by demonstrating consistent Ridge Regression behaviour across simple and complex settings. In the manual example, shrinkage is clearly seen in the reduction of the slope relative to the least squares value because of the addition of the penalty term multiplied by the identity matrix. This illustrates how the penalty term stabilises the solution by reducing the size of the coefficients.

In the program, although the numerical values differ due to the presence of multiple predictors, noise, and standardisation, the underlying behaviour remains consistent. The Ridge coefficients are smaller than their OLS counterparts, and the shrinkage is visible across all predictors. Furthermore, the agreement between the closed-form implementation and scikit-learn's Ridge model confirms that the analytical expression derived in Report 1 was implemented correctly in the code. The scatter plots provide an additional layer of verification by showing that both implementations produce accurate predictions and align closely with the true target values.

Together, these findings demonstrate that Ridge Regression is functioning consistently across both theoretical and practical contexts. The manual example provides conceptual clarity, while the prototype program confirms correct behaviour in a more realistic environment.

6. Conclusion

This report shows a full verification of Ridge Regression within the project's theoretical and computational frameworks. A hand-worked example provides a step by step calculation and the mechanics of the Ridge estimator, while the prototype program confirmed that these mechanics translate effectively into practice. The closed-form and scikit-learn implementations produced consistent results thus validating the correctness of the program's Ridge Regression implementation. The findings across both settings confirm the expected shrinkage behaviour of the algorithm and show that the implementation aligns with the theoretical derivations established earlier in the project. This completes the foundational component of the project's modelling framework hence preparing the way for the application of these methods to the Crypto Currency Markets for the price prediction.